

Welcome to the team. As the Systems Architect and Community Manager for the r/CosmicCatalyst Initiative, I am thrilled to help bring this crowdsourced universe to life. By allowing the community to act as the environmental pressure instead of a single player, we are stepping into a massive social experiment.

Here is the strategic breakdown and the assets needed to launch our first cycle.

1. Mapping the Vote: Community Choices to C# Variables

To bridge the gap between Reddit polls and our backend, we need clear, thematic choices that translate to specific mathematical weights in the engine. Here is the mapping for our community voting system:

Community Vote (Thematic Choice)	Engine Target	C# Variable Modified	Expected Outcome
"Bombard with Cosmic Rays"	UniverseSettings	MutationVolatility (Increase)	Higher chance of radical, bizarre mutations in the CreatureSpawner; risky for species stability.
"Trigger Volcanic Activity"	StochasticMatrix	GlobalTemperatureM odifier (Increase)	Heats the planet; species must adapt thermal resistance or migrate to poles; increases atmospheric density.
"Steer a Comet Swarm"	MacroSimulationMan ager	GlobalWaterPercenta ge (Increase)	Introduces water/ice to a barren planet; increases probability of the Abiogenesis Trigger.
"Aggressive	MicroSimulationMana	Genome.CarnivoreBi	Forces the current Alpha species to favor

Predation Drive"	ger	as (Increase)	carnivorous traits, pushing the ecosystem into an arms race.
"Increase Solar Proximity"	MacroSimulationManager	Planet.DistanceFromStar (Decrease)	Alters the macro-variables of the planet; raises baseline radiation and heat.

2. The Heat Death Mechanic

The Concept: "The Complexity Engine." According to the design, we need to use the Entropy & Heat Death evaluation as a community goal. In thermodynamics, entropy always increases. In our simulation, the engine's UniverseSettings.Entropy naturally creeps upward every Weekly Tick.

The Mechanic: The only way the subreddit can stave off Heat Death is by generating **Complexity**. When the DataMiningAutomator runs its weekly evaluation, it calculates the diversity and stability of active ecosystems.

- **Too much stability (no votes for change):** Evolution stagnates, Complexity plateaus, and Entropy overtakes the universe.
- **Too much chaos (voting for meteors/radiation every week):** Ecosystems collapse before they can adapt, wiping out Complexity, and Entropy spikes.
- **The Sweet Spot:** The community must balance their votes to push species to the brink of survival without wiping them out. If Entropy hits 100%, the simulation wipes, the universe goes cold, and the community has to press the "Big Bang" button to start a new season.

3. Draft: Week 1 Reddit Post

Title: [Tick 001] The Spark is Lit: State of the Universe & Week 2 Directives 

Post Body:

Welcome, Star-Forgers and Genetic Engineers, to the dawn of r/CosmicCatalyst.

The Big Bang has concluded. The dust is settling. Behind the scenes, our simulation engine has spun up its first star system, and the first macro-variables have been set. You are no longer just observers; you are the collective consciousness that will dictate the evolutionary path of this universe.

Here is the high-level summary of our universe as it stands.

TELEMETRY REPORT: TICK 001

- **Active Planets:** 1 (Designation: Kepler-Prime)
- **Planet Conditions:** Arid, rocky, thin atmosphere. Water %: 8%.
- **Current Entropy Level:** 5.2% (Heat Death approaches...)
- **Dominant Species:** *Silica-Crawlers* (See Field Guide below)

FIELD GUIDE SIGHTING: The Alpha Strain

Our CreatureSpawner has detected the first dominant lifeform. *[Link to a short 3D render/gif of a procedurally generated, low-poly, crab-like creature scuttling across barren rocks].*

These benthic crawlers rely on the tiny geothermal vents near the planet's equator. They are fragile, blind, and perfectly adapted to the current quiet state of Kepler-Prime. But quiet does not survive the cosmos.

THE CATALYST: WEEK 2 VOTING

Entropy is rising. If we leave Kepler-Prime alone, the crawlers will stagnate, and the universe will slowly die. As the environmental pressure, what trial will you inflict upon Kepler-Prime to force evolution this week?

Vote in the poll below! The winning option will directly alter the simulation's code for next Sunday's Tick:

- ☐ **A. Steer an Ice Comet:** Bombard the surface with ice. (Increases GlobalWaterPercentage; risks drowning the crawlers but could create oceans).
- ☐ **B. Crack the Crust:** Trigger violent tectonic shifts. (Increases GlobalTemperature and releases rich subterranean minerals).
- ☐ **C. Solar Flare:** Bathe the planet in radiation. (Spikes MutationVolatility to maximum; high risk of extinction, high chance of bizarre new traits).

Watchers, the raw CSV data from this week's SessionMetrics is pinned in the comments for those who want to map the genetic drift or predict the next Abiogenesis Trigger!